

2025 AMC 10A Problem 8 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 8

Problem:

Agnes writes the following four statements on a blank piece of paper.

- At least one of these statements is true.
- At least two of these statements are true.
- At least two of these statements are false.
- At least one of these statements is false.

Each statement is either true or false. How many false statements did Agnes write on the paper?

Answer Choices:

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

Solution:

Let the four statements be S_1, S_2, S_3, S_4 in order, and let T be the number of true statements and F the number of false ones, so $T + F = 4$.

- If $T = 0$, then all are false. But then S_3 says "at least two of these statements are false," which would be true. Contradiction. So $T \neq 0$.
- If $T = 4$, then all are true. But then S_3 ("at least two are false") and S_4 ("at least one is false") would both be false. Contradiction.
- If $T = 2$, then $F = 2$. Then S_3 ("at least two are false") is true, and S_4 ("at least one is false") is also true, and S_1 ("at least one is true") is true as well. That gives at least three true statements, so $T \geq 3$, a contradiction.
- If $T = 1$, then $F = 3$. Then S_1 ("at least one is true") must be true. Since $F = 3$, S_3 ("at least two are false") is also true, giving at least two true statements, so $T \geq 2$, a contradiction.

The only remaining possibility is

$$T = 3, \quad F = 1.$$

Indeed, if three statements are true and one is false, then S_1, S_2 , and S_4 are true, and S_3 is false, which is consistent.

Therefore, the number of false statements Agnes wrote is

(B) 1

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